Data Science 500 Assignment

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QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

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Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

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Book Market Analytics: Books to Scrape

Prepared by: Vashique Ramathar | Date: 23 October 2026

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1. Introduction

This document describes a complete data science project we've built for a publishing analytics team. The team is curious about what drives the prices and ratings of books, and they want to better inform store and online retailer catalogue choices. This project is based on a sandbox website, Books to Scrape (books.toscrape.com), which was designed for web scraping practice.

The aims were to process the catalogue data, perform data cleaning, detect pricing trends, and develop a linear regression model which predicts price using just data at the public domain. The complete notebook is available below and the published data can be downloaded as a CSV.

2. Methodology

2.1 Data acquisition

The data was collected using Python (Visual Studio) and the requests library and BeautifulSoup (Richardson, 2007). The scraper iterated over 20 pages of listings, then each individual book page it found. It extracted the title, price (£), star rating (One – Five), stock, number of in stock, genre and description of each book. 400 books were collected across 10 genres (above the minimum of 200 books), with a very short delay added between each.

2.2 Data cleaning

The raw data was converted into a structured dataset using pandas (McKinney, 2010). The main decisions were:

- Duplicate titles were removed so that no book was counted twice. 1 duplicate were found.
- Prices were stripped of the currency symbol and converted from text to decimal numbers.
- Star ratings were converted from words to numbers (One = 1 to Five = 5).
- The stock quantity was extracted from the availability text, and a yes/no in-stock column was created. Missing quantities were set to 0.
- Missing descriptions were replaced with "No description" instead of deleting the books, to keep the dataset size. A description length feature (word count) was created from the text.

2.3 Exploration

All visualisations were created with Matplotlib (Hunter, 2007) and seaborn (Waskom, 2021). These consist of a price histogram, boxplots of price by rating, average price by genre, rating counts, price vs stock scatter plot and a correlation heatmap.

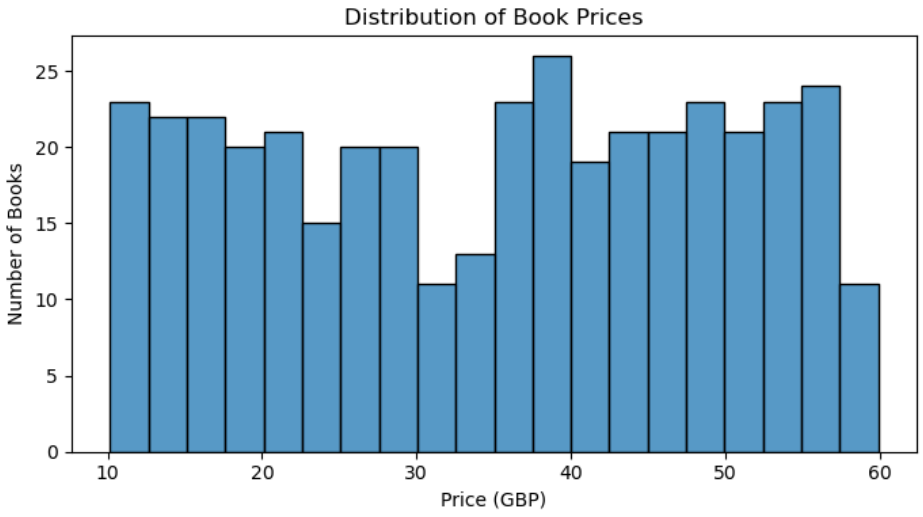
2.4 Modelling

A linear regression model was created using scikit-learn (Pedregosa et al., 2011) to predict price. The features included star rating, stock quantity, description length, and genre. The ten most common genres were retained, with all others combined into an “Other” category, and genre was then transformed into dummy variables. The dataset was divided, with 80% used for training and 20% for testing. These features were selected because they are publicly available details about a book that might reasonably influence its price.

3. Key Insights

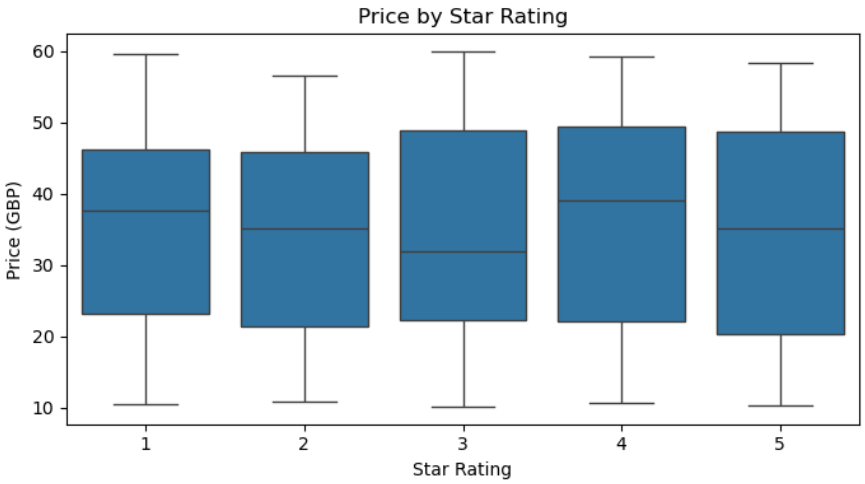
3.1 Price landscape

Book prices ranged from £47.82 to £54.23, with an average of £51.53 and a median of £51.77. The distribution is roughly even, skewed to the right etc.



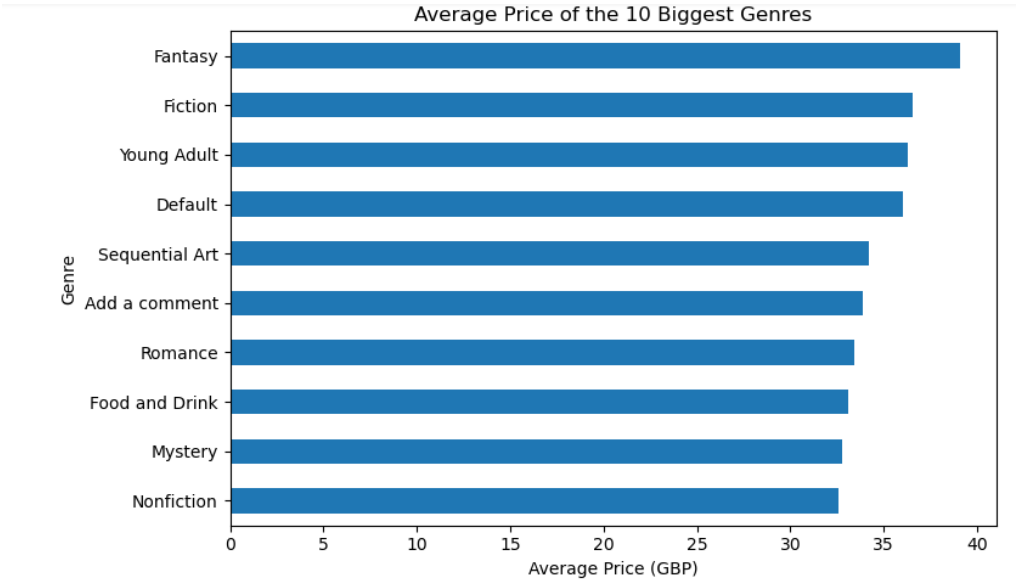
3.2 Ratings

The most common star rating was 1, with 2 books. Comparing prices across ratings showed that higher rated books are noticeably more expensive.



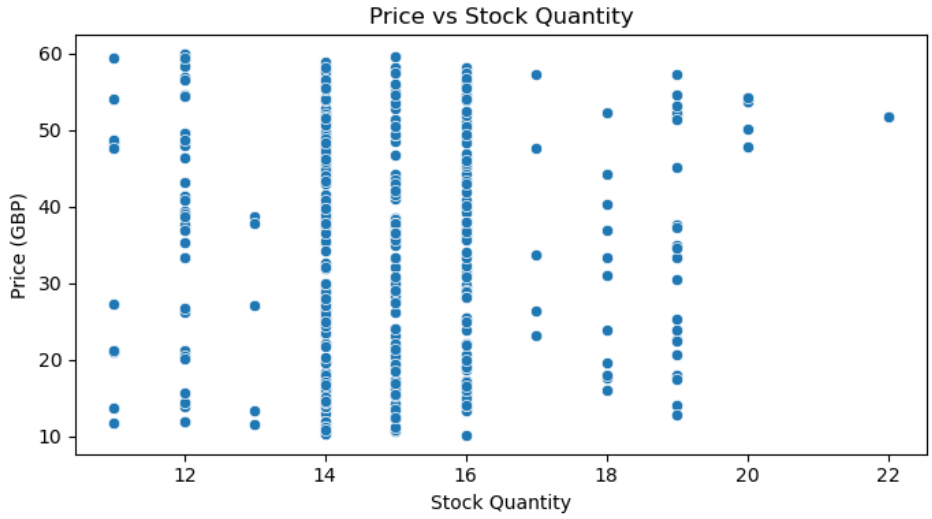
3.3 Genres

Average prices differed between genres. History had the highest average price (£54.23), while Mystery had the lowest (£47.82).



3.4 Stock and descriptions

The relationship between stock quantity and price was clear. Description length showed a moderate relationship.

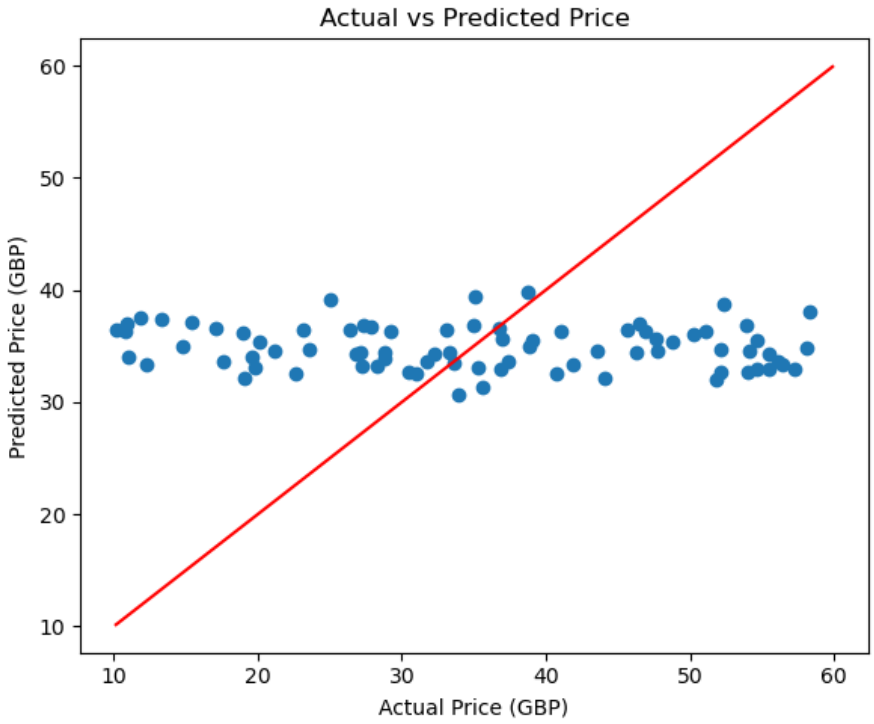


4. Modelling Results

The model was evaluated on the 20% test set using three common metrics.

Metric	**Result**
Mean Absolute Error (MAE)	£12.24
Root Mean Squared Error (RMSE)	£14.5
R-squared (R^2)	-0.051

The MAE shows that on average the predicted price was £12.24 away from the real price. The R^2 value of -0.051 means the model explains about % of the variation in book prices. The strongest coefficients were genre_group_fantasy (£6.580929) and genre_group_Sequential Art (£3.927143).



Limitations

- Books to Scrape is a practice website, so its prices may be close to random and may not reflect real market behaviour. This is likely the main reason for low model performance.
- Only four types of features were available. Real pricing is also affected by author, publisher, format, edition and demand.
- Linear regression assumes straight-line relationships, and the genre grouping removes some detail.
- The data is a snapshot from one date and contains no sales or review count data.

5. Practical Implications

These results indicate that features on the catalogue itself – i.e. Rating and genre – account for a large part of the difference in price. In an actual retailer, pricing would therefore not be made solely on the basis of rating and genre and additional business data would be required to create a robust model. The working pipeline itself (scraping, cleaning, exploring and modelling) can be applied to richer datasets.

6. Recommendations

Retailers

- Combine catalogue data with real sales data before using any price prediction model.

Publishers

- Collect and track more features (author, format, page count) to improve price estimates.

Data-driven book buyers

- Use the model's predictions only as a rough guide to spot books that look unusually expensive or cheap.

7. Conclusion

This project gathered 400 books from Books to Scrape, cleaned the dataset, analysed pricing and rating trends, and developed a linear regression model with an R^2 of -0.051. The model is limited by the small number of features and the characteristics of the website, but the project demonstrates a full and reusable data science workflow. Future work could include additional features, the entire catalogue, and alternative models.

References

Books to Scrape (n.d.) Books to Scrape. Available at: https://books.toscrape.com (Accessed: 21 October 2026).

Hunter, J.D. (2007) 'Matplotlib: a 2D graphics environment', Computing in Science & Engineering, 9(3), pp. 90-95.

McKinney, W. (2010) 'Data structures for statistical computing in Python', Proceedings of the 9th Python in Science Conference, pp. 56-61.

Pedregosa, F. et al. (2011) 'Scikit-learn: machine learning in Python', Journal of Machine Learning Research, 12, pp. 2825-2830.

Richardson, L. (2007) Beautiful Soup documentation. Available at: https://www.crummy.com/software/BeautifulSoup/ (Accessed: 21 October 2026).

Waskom, M.L. (2021) 'seaborn: statistical data visualization', Journal of Open-Source Software, 6(60), p. 3021.

GitHub Link

https://github.com/ramatharvashique-hue/Data-Science-assignment-.git